

AURIX™ TC3xx Safety Mechanism Hints

Interpretation and Implementation Information

About this document

Scope and purpose

The purpose of this document is to provide interpretation and implementation hints for the AURIX™ TC3xx microcontroller family safety mechanisms and the actions that shall be taken by the product system integrator to ensure the correct operation of the device. The description is limited to those technical actions that are required to ensure compliance with the relevant safety standards.

This document is intended to support the AURIX™ TC3xx Safety Manual [1] and does not replace it or contradict it. Other documents may be referenced for information that is deemed outside the scope of this document. These documents can be found on the Infineon website www.infineon.com, or are available on request.

Intended audience

This document is written for system engineers, software engineers, and functional safety managers involved in the design or development of a safety-related system who are considering integrating the AURIX™ TC3xx microcontroller hardware as a safety element out of context (SEooC) into their system.

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1 ESM[HW]:SYS:WATCHDOG_FUNCTION

Description

An external device with an independent reference clock and with the functionality of a time-window watchdog supervises the AURIX™ microcontroller. This external device must initiate the transition to the system safe state if a fault which can lead to the violation of a system safety goal is detected.

Note: This is necessary to control common cause failures initiated by external stress conditions or internal failures of the microcontroller, which may lead to a state where the microcontroller cannot signal an internal failure.

Some systems implement external watchdogs supporting program flow monitoring with a question-answer protocol. Although these functions can be taken over by the internal watchdogs, cancelling the need for such a device, complex external watchdogs are still supported by the AURIX™ microcontroller. In this case a separate window watchdog may not be required.

Interpretation

This safety mechanism requires the use of an external watchdog device which is able to detect faults in the microcontroller which could lead to the violation of the system's safety goals. The required device shall be able to monitor the behavior and the plausibility of the program sequence executing on the AURIX™ microcontroller.

However it is allowed for the system integrator to use any other device or combination of devices and safety mechanisms provided that the devices used achieve at least the same performance, in terms of covered failure modes, diagnostic coverage, and independency.

A possible implementation could be the use of a timeout watchdog (see ISO26262-5, D.2.9.1) or time-window watchdog (see ISO26262-5, D.2.9.2) in combination with temporal and logical monitoring of program sequence (see ISO26262-5, D.2.9.3). Such a combination is described in ISO26262-5 D.2.9.4.

2 ESM[HW]:PMS:VEXT_VEVRSB_OVERVOLTAGE

Description

It is assumed that the system detects and disables the external voltage supplies VEXT and VEVRSB of the microcontroller in case of an over-voltage condition, through the continuous monitoring of the external voltage supplies of the microcontroller.

The over-voltage can be divided into 2 regions:

- Up to absolute maximum rating
 - Mechanisms are in a separate domain. Exceeding the operating conditions for longer than the specified time may lead to an increase of the microcontroller failure rate
- Above absolute maximum rating
 - This is the reliability issue.
 - In case internal circuitry is damaged, LBIST and HWBIST will detect it during start-up
 - If the microcontroller is permanently damaged due to the exposure to the voltage level exceeding the specified maximum supply voltage, self-test (LBIST, HW BIST) could detect this fault during start-up

Note: Please refer to AURIX™ TC3xx User Manual v1.3.0 [1], chapter 11.2.2.5 [2] and the device datasheet "Operating conditions" chapter for more details about Supply Voltage Monitoring.

Interpretation

This safety mechanism can be decomposed in to two parts:

1. Detection of the over-voltage condition of external voltage supplies VEXT and VEVRSB
2. Avoidance of the propagation of the over-voltage fault which could lead to a critical fault at microcontroller level and eventually to the violation of a safety goal at system level

The first part of the safety mechanism shall be fulfilled through the continuous monitoring of the external voltage supplies VEXT and VEVRSB of the microcontroller.

The second part of the safety mechanism can be fulfilled by disabling the external voltage supplies VEXT and VEVRSB.

However, if the system integrator can ensure in their application that, with other safety mechanisms, no system level safety goal will be violated in case an over-voltage condition of external voltage supplies VEXT and VEVRSB appears, it is not mandatory for them to disable the external voltage supplies VEXT and VEVRSB.

3 SMC[SW]:DTS:DTS_CFG

Description

The application software shall configure die temperature limits for the upper and lower thresholds in all DTS instances.

Note: This SMC is related to SM[HW]:DTS:TEMPERATURE_MONITOR.

Please refer to AURIX™ TC3xx User Manual v1.3.0, chapter 11.2.1.4 [2] and device datasheet "Temperature sensor" chapter for more details about the DTS.

Interpretation

If the system integrator can ensure in their application through an external temperature sensor or other system level safety mechanisms that the microcontroller's junction temperature, T_j , is always in the specified range, or that the upper/lower temperature limit will never be violated in their application, there is no need to react on the alarms triggered by the violation of the configured temperature limits for the upper or lower thresholds in the DTS instances.

References

4 References

- [1] AURIX™ TC3xx Safety Manual v1.05
- [2] AURIX™ TC3xx User Manual v1.3.0

Revision history

Document version	Date of release	Description of changes
V1.0	2019-10-31	Initial version

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